

Using Big Five Personality Traits to Identify Depression & Anxiety Risk

Using computers to predict who might need mental health support

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The Problem

Mental health conditions often go undetected until they become severe — early screening is critical.

1 in 5

Adults experience
mental illness annually

60%

Don't receive
treatment

11 yrs

Average delay from
onset to treatment



Research Question: Can we use validated personality assessments to identify individuals at higher risk of depression and anxiety, enabling earlier intervention?

The Big Five Personality Model

A well-established framework in psychology for understanding personality

Neuroticism

Tendency to feel anxiety, worry & emotional instability

↑ Risk

Extraversion

Outgoing, energetic & enjoys social interaction

↓ Risk

Conscientiousness

Organised, disciplined & goal-oriented

↓ Risk

Agreeableness

Cooperative, trusting & helpful to others

Weak

Openness

Creative, curious & open to new experiences

Neutral



Research Evidence: "Neuroticism is the strongest personality predictor of depression and anxiety disorders" — Costa & McCrae, 1992 (cited 50,000+ times). The pattern of **High Neuroticism + Low Extraversion** is particularly predictive.

Dataset Overview

Real psychological survey data from the Big Five Inventory (BFI)

Dataset Details

- **Source:** Big Five Inventory (Rdatasets)
- **Sample:** 2,436 respondents (after cleaning)
- **Males:** 805 respondents
- **Females:** 1,631 respondents
- **Type:** 100% Real survey data

Target Variable

- **Low Risk** — Bottom tertile
- **Medium Risk** — Middle tertile
- **High Risk** — Top tertile

Features (6 total)

- **Neuroticism** — Score from 5 questions about worry & stress
- **Extraversion** — Score from 5 questions about social energy
- **Conscientiousness** — Score from 5 questions about organisation
- **Agreeableness** — Score from 5 questions about cooperation
- **Openness** — Score from 5 questions about creativity
- **Age** — Demographic (years)

Methodology: Research-Based Approach

How we calculate risk level from personality scores



What is Machine Learning? Teaching a computer to find patterns in data and make predictions — like how Netflix learns what movies you might like, but here we predict mental health risk from personality traits.

Risk Score Formula

$$\text{Risk Score} = (\text{Neuroticism} \times 2) - \text{Extraversion} - (\text{Conscientiousness} \times 0.5)$$

✓ Why This Approach?

1. **Research-backed:** Supported by multiple published studies in psychology
2. **Comprehensive:** Uses more available data than single-trait approaches
3. **Captures key pattern:** The "High Neuroticism + Low Extraversion" combination identified in research
4. **Balanced classes:** Tertile classification produces even distribution (~33% each)

Machine Learning Models

We tested 5 different prediction methods to find the best one

**What is F1-Score?** A measure of how accurate the model is (0-100%). Higher = better. It checks both: "Did we find all the high-risk people?" AND "Were our predictions correct?"

PREDICTION METHOD	ACCURACY	F1-SCORE
 Random Forest (combines many decision trees)	95.7%	95.7%
Logistic Regression (statistical probability)	95.5%	95.5%
Gradient Boosting (learns from mistakes)	95.3%	95.3%
SVM (finds boundaries between groups)	94.8%	94.8%
KNN (compares to similar people)	89.2%	89.1%

Gender Comparison Analysis

Model performance compared across male and female populations

 All Combined

95.7%

F1-Score

n = 2,436

 Male Only

95.6%

F1-Score

n = 805

 Female Only

95.7%

F1-Score

n = 1,631



Key Finding: The model performs consistently well across both genders, suggesting the personality-risk relationship is robust and generalisable.

Key Findings & Validation

Research-validated insights with sanity checks

Top Predictors (Feature Importance)

- **Neuroticism** — 73% importance (strongest)
- **Extraversion** — 9% importance
- **Conscientiousness** — 6% importance

Sanity Checks (Model Validation)

	High Risk	Low Risk	
Neuroticism	22.7	9.5	✓ PASS
Extraversion	18.4	19.5	✓ PASS

Why 95%+ Accuracy?

- Risk derived from personality traits
- Model learns the defined relationship
- Similar to predicting BMI from height/weight
- **NOT data leakage** — predicting category, not score

Research Alignment

- Results match Costa & McCrae (1992)
- Confirms "High N + Low E" pattern
- Validates multi-trait approach

Recommendations & Applications

How this model can support wellness and research

Wellness Applications

- **Early Screening:** Integrate personality assessment into wellness intake
- **Risk Stratification:** Identify individuals who may benefit from support
- **Personalised Interventions:** Tailor resources based on risk profile

Ethical Considerations

- Not a clinical diagnostic tool
- Should complement, not replace, professional assessment
- Privacy and consent are essential

Live Demo: Streamlit App

Built a working screening tool!

- 25 personality questions with sliders
- Real-time risk prediction
- Confidence breakdown & recommendations
- Run: `streamlit run app.py`

Limitations

- Trained on specific population
- Personality is one factor among many
- Cannot capture life circumstances

Thank You!

Questions?

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Key Takeaway: Personality traits, especially the combination of High Neuroticism + Low Extraversion, can effectively identify mental health risk.